In the SI system, the ***mole (mol)* is *the amount of a substance that contains as many elementary entities (atoms, molecules, or other particles) as there are atoms in exactly 12 g (or 0.012 kg) of the carbon-12 isotope.*** The actual number of atoms in 12 g of carbon-12 is determined experimentally. This is called **Avogadro's Number (N_A),** named after an Italian scientist **Amedeo Avogadro** who proposed its significance. Its value is **6.023×10^{23}**. So one mole of a substance contains **6.023×10^{23}** entities. Thus, 5 moles of oxygen molecules contain $5 \times 6.023 \times 10^{23}$ molecules.

Mole Concept: The study of the collection of particles by using mole as the counting unit, in order to express the mass and volume of such unit particles in a bulk of matter is known as **mole concept**.

The number of moles of a substance can be calculated by various means depending on the data available, as follows:

- Number of moles of molecules.
- Number of moles of atoms.
- Number of moles of a gas (Standard molar volume at STP = 22.4 litre).
- Number of moles of ions.

Note:

STP-Standard Temperature and Pressure(273.15 K,1.00 atm)

Mole of atoms:

One mole of an element contains 6.023×10^{23} atoms and it is equal to its gram atomic mass.

i.e., one mole of oxygen atom contains 6.023×10^{23} atoms of oxygen and its gram atomic mass is 16 g.

Mole of molecules:

One mole of matter contains 6.023×10^{23} molecules and it is equal to its gram molecular mass.

i.e., one mole of oxygen molecule contains 6.023×10^{23} molecules of oxygen and its gram molecular mass is 32 g.

Molar volume:

One mole of any gas occupies 22.4 litre or 22400 ml at S.T.P. This volume is called as molar volume.

Calculation of number of moles by Different modes

Number of moles

= Mass / Atomic Mass

= Mass / Molecular mass

= Number of Atoms / 6.023×10^{23}

= Number of Molecules / 6.023×10^{23}

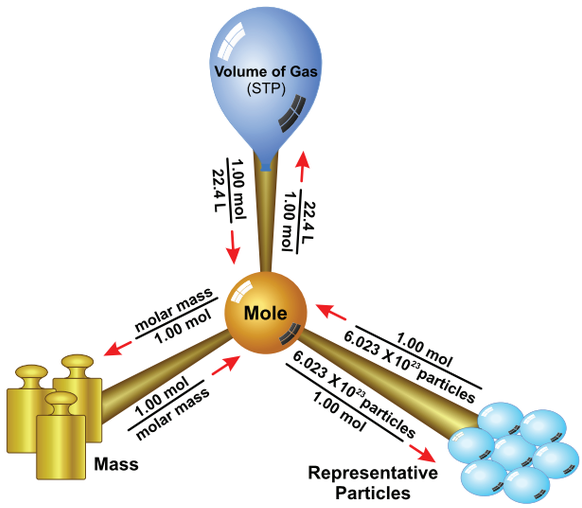


Figure 7.3 Mole concept

7.5 PERCENT COMPOSITION

So for, we were dealing with the number of entities present in a given substance. But many times, the information regarding the percentage of a particular element present in a compound is required.

The percentage composition of a compound represents the mass of each element present in 100 g of the compound.

Let us understand the percentage composition of oxygen and hydrogen by taking

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